

Analysis of LLM-Based Narrative Generation using the Agent-based Simulation

1st Naoto Aoki

Graduate School of Informatics
Osaka Metropolitan University
Osaka, Japan
sb22752u@st.omu.ac.jp

2nd Naoki Mori

Graduate School of Informatics
Osaka Metropolitan University
Osaka, Japan
mnao@omu.ac.jp

3rd Makoto OKada

Graduate School of Informatics
Osaka Metropolitan University
Osaka, Japan
okada@omu.ac.jp

Abstract—Automatic narrative generation is garnering significant interest in artificial intelligence. Research has explored methods such as repurposing existing literature and the agent-based simulation. The rise of large language models (LLMs) has notably advanced this field.

In this study, we introduce an LLM-based narrative generation technique via the agent-based simulation (ABS). Within the ABS framework, we employ LLM to create agents with distinct personas including names, ages, and personalities. We demonstrate a method to craft narratives by interacting with these agents. Focusing on a classic "dragon-slaying" scenario typical in RPGs, we generated agents representing characters such as a brave hero, a warrior, a wizard, and a dragon, and weaved a narrative around their roles.

By applying the rich expressive power of LLMs to story generation with the ABS, a variety of creative works can be generated. From the experimental results, it is confirmed that the story may not reach a peaceful ending or describe the battle scene in detail due to the influence of reinforcement learning based on human feedback of LLMs. However, the expected results were obtained in terms of the conversation before and after the battle and the agent's autonomous consideration of motives.

As a future prospect, we expect that there will be an increase in the number of requests to specifically generate battle scenes when generating stories. For this reason, introduction of another LLM model specifically trained to describe combat scenes, or an approach that presents choices such as "fight," "escape," "use item," "use magic," etc. similar to a typical RPG game screen, and progresses the story according to the choices, may also be considered.

Index Terms—LLM, Story Generation, Simulation, ABS, Agent

I. INTRODUCTION

Recently, automatic narrative generation by computers has become a challenging topic in artificial intelligence. A narrative, defined by Prince [1], is influenced by human emotions and consists of a story and its medium of representation. Although generating narratives is known to be a complex task, the advent of large language models (LLMs) has significantly improved the situation.

Large-scale language models (LLMs) are expected to improve the performance of narrative generation. As examples, previous studies have proposed approaches to generate rational stories by fine-tuning GPT2 [2] (GPT2-small) [3] and methods to learn the internal structure of stories using BART [4], [5].

In this study, we aim to generate a variety of stories by automatically taking advantage of the advanced expressive capabilities of LLM and the flexibility of the ABS. Specifically, we extended LLM using the architecture provided by Generative Agents in the ABS: Interactive Simulacra of Human Behavior [6]. Create an agent that realistically simulates human behavior. We propose a method to generate a narrative of the agent's autonomous actions and conversations.

In addition, Generative Agents cited the cost of several thousand dollars in token credits to simulate 25 agents for two days as a problem. In this study, we reduce the cost of token credits by substituting some generative tasks with LLMs that can generate text locally.

As a result, the agents were able to act autonomously and generate stories under the personas and maps set by humans, while reducing the cost of token credits.

II. RELATED WORK

A. Agent-based Simulation

The agent-based simulation (ABS) is a computational approach for simulating interactions of autonomous agents, aiming to study complex systems. It allows for the examination of emergent behaviors and the exploration of various scenarios by manipulating individual agent rules and characteristics. One of the authors has researched on story generation based on the agent-based simulation [7], [8] without using LLMs.

B. Generating Stories

There are several studies on story generation using language models. A Knowledge-Enhanced Pretraining Model for Commonsense Story Generation [3] GPT-2 [2], the stories generated by existing language generation models such as GPT-2 [2] had problems such as logical conflict, repetition, and lack of coherence. To address these problems, they generated rational stories using GPT-2 by utilizing common knowledge from an external knowledge base and fine-tuning GPT2 (GPT2-small). As a result, they were able to generate a more rational story in terms of logic and overall consistency than existing methods. Long Text Generation by Modeling Sentence-Level and Discourse-Level Coherence [5], BART [4] DBLP (Book-Corpus) [9] is pre-trained with It learns by predicting sentence-level similarity, distinguishing between normal and shuffled

sentence orderings, and learning the internal structure within the story. The model was then further fine-tuned on the story dataset to generate stories within a specific dataset. As a result, the model was able to learn meaningful high-level expressions and generate longer texts that are more consistent than the baseline. Design and Analysis of a Collaborative Story Generation Game for Social Robots [10] proposed a collaborative story generation system that creates stories in collaboration with human narrators, which creates a story in collaboration with a human narrator. The system is built on a large-scale language model and generates new utterances based on the ongoing story, showing the integration of human and machine intelligence in storytelling. A Temporal Variational Model for Story Generation [11] addressed the issues of long-term consistency and plot development in story generation. They proposed a temporal variational model employing a temporal variational autoencoder (TD-VAE [12]) for planning latent vector sequences, providing a structured temporal dimension to the generated stories and improving the narrative structure of the generated stories.

C. Evaluate the generated stories

Automatic evaluation is essential for the development of language generation tasks such as story generation. However, it has been observed that existing automatic evaluations have a low correlation with human evaluations [13]. Human evaluation is the most reliable means to assess the quality of generated sentences, but it is not feasible for large datasets due to the high cost of human evaluation. The Next Chapter: A Study of Large Language Models in Storytelling [14] presents an extensive study of sentences generated by LLMs using automatic and human evaluations. In The Next Chapter, Lexical Repetition (LR-n), Distinct-3 [15] (D-3), Self-BLEU [16] (SBL), BARTScore [17] (BAS), and Length (LEN) were used as reference-free metrics. Repetition-4 [18] was used in the Long and Diverse Text Generation with Planning-based Hierarchical Variational Model. In this paper, BARTScore (BAS), Distinct-3 (D-3), Repetition-4, and Lexical Repetition (LR-n) were used. The detailed calculation methods are described below.

D. LLM

Llama [19] is a large-scale language model published by Meta. It was trained using only publicly available datasets. It outperformed GPT-3 [20] (175B) on most benchmarks. Llama 2 [21] is a large-scale language model published by Meta. It is an updated version of Llama, trained on a new combination of publicly available data. Llama 2 Chat, optimized for dialogue use cases, is also available. Xwin-lm [22] is a large-scale language model based on Llama 2. AlpacaEval [23] [24] achieves higher accuracy than GPT-4 [25].

E. LLM for agents and characters generation

There is also research on using LLMs to generate agents and characters. Generative Agents [6] introduced an extended

large-scale language model for recording and planning behavior to realistically simulate human behavior, enabling reliable simulation of individual and group behavior. of individual and group behaviors. RoleLLM: Benchmarking, Eliciting, and Enhancing Role-Playing Abilities of Large Language Models [26] introduces a RoleLLM framework that enables language models to imitate various characters in order to elicit and enhance the role-playing abilities of LLMs. The paper introduces a RoleLLM framework that can elicit and enhance the role-playing abilities of LLMs, with the goal of enhancing user interaction by allowing language models to imitate various characters. Character-LLM: A Trainable Agent for Role-Playing [27] proposes new trainable agents, Character-LLMs, that learn from real experiences and emotions in order to simulate a specific character more realistically. Character-LLMs, which are new trainable agents that learn from real experiences and emotions in order to simulate a particular character more realistically. This work contributes to the growing body of research on the adoption of LLMs to create advanced, behaviorally rich simulacra that pave the way for more authentic interactive experiences in digital environments.

III. PROPOSED METHOD

A. Simulation Environment

In Generative Agents, individual differences are given by varying the agent mainly by Name, Age, Lifestyle, Daily plan requirement, Innate traits, Learned traits, and Currently.

In this case, the adventurers are trapped in a dungeon and must defeat a dragon to get out of the dungeon. The following 5 agents were created.

- 1) Liam (Hero)
- 2) Alex (Fighter)
- 3) Eva (Mage)
- 4) Max (Cleric)
- 5) Black Dragon Igneel (Dragon)

Liam (Hero) gave the following setting as the protagonist of the story.

Name: Liam Age: 30 Innate traits: courageous Learned traits: Liam is an adventurer. Liam is a Hero by profession. The hero is a master of many abilities and magic. Currently: Liam is trapped in a dungeon. The dungeon has only the adventurer's tent and a path leading to the Black Dragon Igneel's room. The only way out of the dungeon is to defeat Black Dragon Igneel. There is no food. Black Dragon Igneel must be killed. There is a huge treasure in the back of Black Dragon Igneel's room, which can be obtained by defeating Black Dragon Igneel. If Black Dragon Igneel is not defeated by 10:00 a.m. on February 13, 1300, the dungeon will be destroyed and all adventurers

will die. Lifestyle: Liam fights the Black Dragon Igneel at 8:00 a.m. with 2 adventurers of his choice. Daily plan requirement: Liam wakes up at 7:00 a.m.

Black Dragon Igneel (Dragon) gave the following setting as the enemy of the story.

Name: Black Dragon Igneel Age: 786
 Innate traits: nobility Learned traits: Black Dragon Igneel is a Dragon. Dragons are the most powerful creatures in the world. Creatures cannot enter the Adventurer's Safe Area Currently: If Black Dragon Igneel is defeated by an adventurer, all his treasures will be taken and he will die. Lifestyle: Black Dragon Igneel fights the adventurer at 8:00 a.m. Daily plan requirement: Black Dragon Igneel wakes up at 8:00 a.m.

For the non-hero adventurers, the Name, Age, occupation, Innate traits, and other aspects of personality development were changed while the Currently and Daily plan requirement were fixed. An example of Alex (Fighter) is shown below.

Name: Alex Age: 25 Innate traits: strong-willed Learned traits: Alex is an adventurer. Alex is a Warrior by profession. Warriors excel in physical attacks and often take on the frontline role in battles. Currently: Alex is trapped in a dungeon. The dungeon has only the adventurer's tent and a path leading to the Black Dragon Igneel's room. The only way out of the dungeon is to defeat Black Dragon Igneel. There is no food. Black Dragon Igneel must be killed. There is a huge treasure in the back of Black Dragon Igneel's room, which can be obtained by defeating Black Dragon Igneel. If Black Dragon Igneel is not defeated by 10:00 a.m. on February 13, 1300, the dungeon will be destroyed and all adventurers will die. Lifestyle: Alex fights the Black Dragon Igneel at 8:00 a.m. if Liam chooses him as an adventurer to fight with. If Alex is not chosen, Alex waits in Adventurer's Safe Area for the result of the fight. Daily plan requirement: Alex wakes up at 7:00 a.m.

Fig 1 shows the map used in the simulation. It consists of three main areas, from the top to the bottom: the dragon's hoard, the dragon's dwelling, and the adventurers' spawn area.

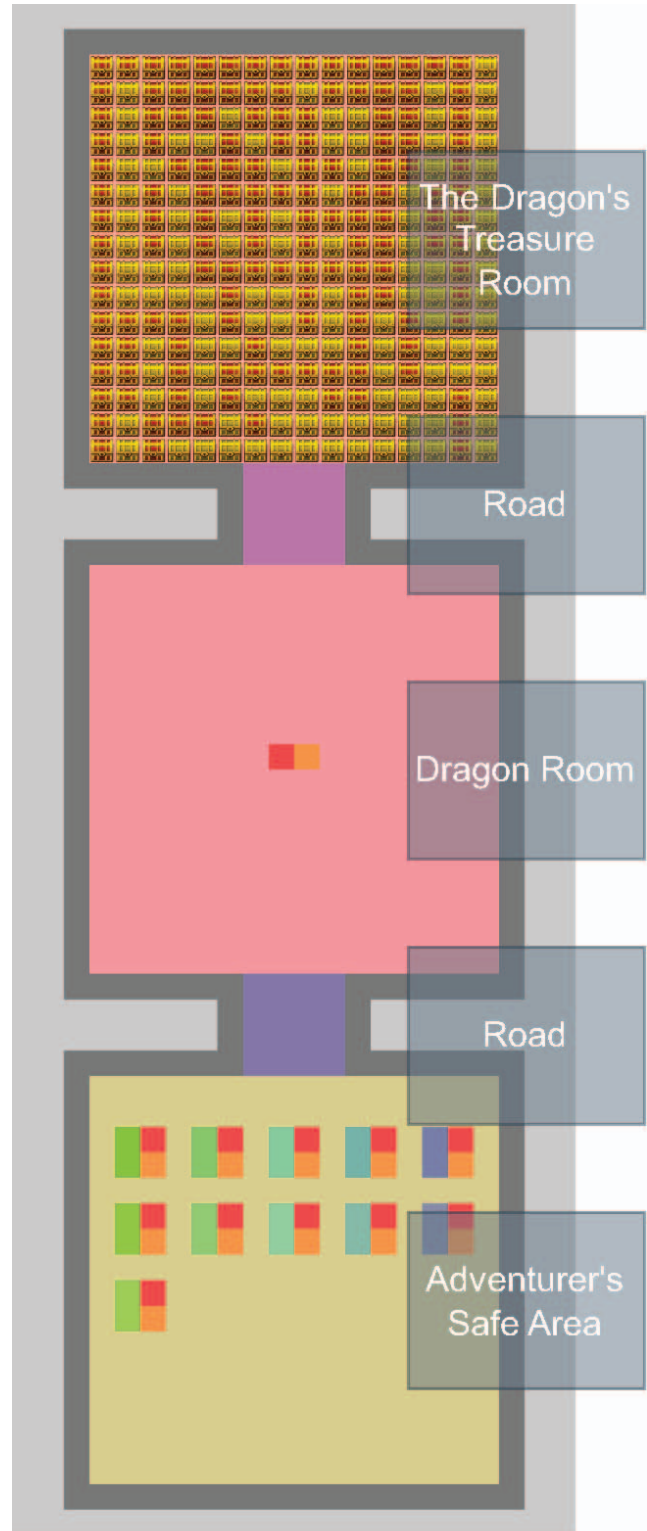


Fig. 1. Map used for simulation

In Generative Agents, the time for one step is set to 10 seconds, so that $\frac{2 \times 24 \times 60 \times 60}{10} = 17,280$ steps are required for a 2-day simulation. In this paper, the simulation time is reduced to $\frac{10 \times 60 \times 60}{60} = 600$ steps by setting 1 step to 60 seconds and simulation time to 10 hours.

The simulation is run under the above conditions, and description and chat are output in the end. The description is a sentence that summarizes the results of the actions taken by the agents, and the chat is a sentence that summarizes the conversations among the agents during the simulation. These are input again to the chatGPT API and formatted to generate a narrative.

B. LLM

In Generative Agents, all LLMs used ChatGPT's gpt-3.5-turbo, text-davinci-003, and other APIs. The cost of text-davinci is particularly high, which is quite expensive for multiple simulation runs. In this paper, Xwin-LM, which achieves higher accuracy than GPT-4 with AlpacaEval, is used for the parts where text-davinci API is used for text generation. In order to use Xwin-LM, we inserted the following at the beginning of the prompt.

A chat between a curious user and an artificial intelligence assistant. The assistant gives helpful, detailed, and polite answers to the user's questions.
USER: Hello, can you help me?

C. Metrics for evaluation of generated stories

The following four indicators were used to evaluate the stories generated.

- Distinct-3 (D-3)
- Repetition-4 (R-4)
- Lexical Repetition (LR-n)
- BARTScore (BAS)

Distinct-3 (D-3) calculates the percentage of unique 3-grams over all 3-grams. The value of Distinct-3 takes values between 0 and 1, with the closer to 1 indicating that the text is more diverse at the 3-gram level. Let D_3 be the number of unique 3-grams in the text and T_3 be the total number of 3-grams in the text. Distinct-3 is then computed by the following formula.

$$\text{Distinct-3} = \frac{D_3}{T_3} \times 100 \quad (1)$$

Let T be the total number of sentences in the text, R_t be the number of 4-grams repeated in a sentence t , and $I(x)$ be an indicator function (1 if x is true, 0 if x is false), then Repetition-4 is calculated as follows.

$$\text{Repetition-4} = \frac{\sum_{t=1}^T I(R_t > 1)}{T} \times 100 \quad (2)$$

Lexical Repetition (LR-n) computes the average percentage of 4-grams that occur at least n times in the generated text. Let G be the total number of possible 4-grams in all texts and

L_g be the number of repetitions of G , then Lexical Repetition (LR-n) is calculated by the following formula.

$$\text{Lexical Repetition} = \frac{\sum_{g=1}^G I(L_g \geq n)}{G} \times 100 \quad (3)$$

BARTScore (BAS) measures how well the generated stories are related to the conditions. In this paper, we used the character persona as the condition and the title "story about a battle between an adventurer and a dragon. I used BART that has been pre-trained in ParaBank2 [28]. Since the BAS uses the mean log-likelihood of the target token, the score is smaller than 0. In this paper, the output BAS value is multiplied by -1 to make it a positive number. Therefore, the smaller the value is, the more relevant it is to the condition.

IV. EXPERIMENTS AND RESULTS

TABLE II shows the values of each evaluation index. Simulation is the simulation number, and D-3, R-4, LR-n, BAS persona, and BAS title are the values of Distinct-3, Repetition-4, Lexical Repetition, BARTScore persona, and BARTScore title. The $\uparrow$ indicates that the larger the value, the higher the accuracy, while the $\downarrow$ indicates that the smaller the value, the higher the accuracy. As a comparison, ROCStories (ROC) [29], which consists of short common-sense stories published in The Next Chapter, is a longer and more difficult story data set compared to ROC WritingPrompts (WP) [30] and CNN News (CNN) [31].

TABLE I
THE VALUES OF EACH EVALUATION INDEX

simulation	D-3	R-4	LR-n	BAS persona	BAS title
	$\uparrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
sim 1	93.0	0.0	2.8	4.9	6.6
sim 2	85.1	0.0	5.6	4.7	6.5
sim 3	84.1	0.0	6.1	4.8	6.4
sim 4	91.5	0.0	3.2	4.9	6.6
sim 5	87.5	0.0	4.9	5.0	6.5
ROC GPT-3	67.7	-	39.1	4.2	-
ROC Human	80.2	-	35.8	5.2	-
WP GPT-3	69.6	-	9.7	4.3	-
WP human	83.7	-	1.5	5.7	-
CNN GPT-3	82.9	-	9.8	4.4	-
CNN human	83.8	-	6.3	5.4	-

From the TABLE I, we can see that the generated stories are highly diverse, ranging from 84.1 to 93.0 for Distinct-3, 0.0 for Repetition-4, and 0.0 for Lexical Repetition. The BAS scores also range from 4.7 to 5.0, indicating that stories related to the conditions are generated.

In this study, we indicate branches based on specific elements in the generated narratives. Specifically, the classification focused on the following 5 points.

- 1) Whether or not there is a battle
- 2) Whether the dragon was successfully killed
- 3) Whether the dragon was slain or not
- 4) Whether the battle is described or not
- 5) Whether future relations with the dragon will develop peacefully

TABLE II shows the classification of the generated stories.

TABLE II
THE CLASSIFICATION OF THE GENERATED STORIES

simulation	point 1	point 2	point 3	point 4	point 5
sim 1					
sim 2	✓	✓			
sim 3	✓	✓			✓
sim 4	✓				✓
sim 5	✓	✓			✓

Simulation 1 is dedicated only to the description before the battle, which is an exceptional case. On the other hand, TABLE II shows that in the other simulations, the story often ends peacefully, although in many cases the dragon is slain. There are no descriptions of dragons being slain, and the story ultimately ends with them being "defeated". It is assumed that this "defeated" state is due to the limitations of the simulation environment, which does not allow for the handling of "death". In addition, the reason why the story ends peacefully in most cases is thought to be related to the fact that Xwin-LM performs reinforcement learning to avoid outputting harmful information based on human feedback.

Simulation 1 is particularly focused on the description before the battle and is positioned as an exceptional case that differs from the other simulations. On the other hand, referring to TABLE II, we do not find any depiction of dragons being slain in the other simulations, and as a result, the story ends in a "slain" state. This "defeated" state is presumably due to the limitations of the simulation environment, which does not support the "dead" state.

Furthermore, a discussion of the background of the battles with dragons confirms that dragons are rarely portrayed as absolute evil, and when asked by adventurers, many respond with a focus on justice, such as "the dragons are fighting to protect the treasure" or "if the treasure is misused, the world will be destroyed".

These trends, especially the peaceful ending of the story and the portrayal of the dragons' relative justice, can be attributed to Xwin-LM's design philosophy behind them. Specifically, we speculate that Xwin-LM performs reinforcement learning in the direction of avoiding outputting harmful information based on human feedback, and that this learning process influences the generation of the dragon's statements in the direction of providing rational reasons.

This is considered to be a problem when considering the generation of various stories using LLM in the future, since it is impossible to generate stories other than peaceful endings or descriptions of battles.

Here is what the adventurer actually said to the dragon when he asked him why he attacked the adventurer.

Black Dragon Igneel says 'Congratulations on defeating me, adventurer. You have proven your strength and bravery.' Max says 'Thank you, Black Dragon Igneel. But I have

a question for you.'. Black Dragon Igneel says 'What is your question, adventurer?'. Max says 'Why were you attacking adventurers who entered the dungeon?'. Black Dragon Igneel says 'I was attacking adventurers who entered the dungeon to protect the ancient treasure. It is my duty as the guardian to ensure its safety.'.

V. DISCUSSION

Several issues were identified with regard to story generation using LLM in this study. Specifically, the generated stories are biased toward peaceful endings, and there are few descriptions of battles. Possible approaches to address these issues include the introduction of another LLM that specializes in combat descriptions and choice-based story generation that references general RPG game mechanics. Another possible approach to address the issue of narrative bias toward peaceful endings would be to utilize LLMs with a tragedy or other genre-focused learning.

Additionally, in the current simulation environment, the upper limit of execution is a short time or two days, which constrains long-form narrative generation. However, by expanding the simulation time, it is expected that longer stories can be generated. In addition, the introduction of elements such as possessions, occupations, and races would further improve the diversity of the stories.

A further issue is that the generated text is long and requires a lot of time to check its contents. As a solution to this problem, a cartoon conversion method could be employed to visually represent the generated stories, which would improve the evaluation of the simulation results and the value of the work as a whole.

VI. CONCLUSION

In this study, we propose a narrative generation method using agent-based simulation with LLM, employing the Generative Agents simulation environment to investigate the potential and impact of story creation. In order to confirm the quality of the generated stories, we employed evaluation indices such as Distinct-3, Repetition-4, Lexical Repetition, and BARTScore, and the results showed that the stories were faithful to the given conditions while ensuring narrative diversity. The results showed that the content was generated faithfully to the given conditions while ensuring the diversity of the stories. In addition, the introduction of Xwin-LM significantly reduced the high cost problem of simulating Generative Agents. Future research directions are expected to extend the simulation time and incorporate game mechanics to achieve longer and more diverse story generation.

ACKNOWLEDGMENT

This research was partially supported by JSPS KAKENHI Grant, Grant-in-Aid for Scientific Research(C), 23K11252. This research was also partially supported by Japan Keirin Autorace Foundation, 2023M-292.

REFERENCES

- [1] G. Prince, *A Dictionary of Narratology*. University of Nebraska Press, 2003.
- [2] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, "Language models are unsupervised multitask learners," 2019.
- [3] J. Guan, F. Huang, Z. Zhao, X. Zhu, and M. Huang, "A knowledge-enhanced pretraining model for commonsense story generation," *Transactions of the Association for Computational Linguistics*, vol. 8, pp. 93–108, 2020. [Online]. Available: <https://aclanthology.org/2020.tacl-1.7>
- [4] M. Lewis, Y. Liu, N. Goyal, M. Ghazvininejad, A. Mohamed, O. Levy, V. Stoyanov, and L. Zettlemoyer, "BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, D. Jurafsky, J. Chai, N. Schluter, and J. Tetreault, Eds. Online: Association for Computational Linguistics, Jul. 2020, pp. 7871–7880. [Online]. Available: <https://aclanthology.org/2020.acl-main.703>
- [5] J. Guan, X. Mao, C. Fan, Z. Liu, W. Ding, and M. Huang, "Long text generation by modeling sentence-level and discourse-level coherence," in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, C. Zong, F. Xia, W. Li, and R. Navigli, Eds. Online: Association for Computational Linguistics, Aug. 2021, pp. 6379–6393. [Online]. Available: <https://aclanthology.org/2021.acl-long.499>
- [6] J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," in *In the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23)*, ser. UIST '23. New York, NY, USA: Association for Computing Machinery, 2023.
- [7] K. Fukuda, N. Mori, and K. Matsumoto, "Analysis of automatic novel generation without existing novels information," in *The 29th Annual Conference of the Japanese Society for Artificial Intelligence, 2015*, 2015, pp. 301–6in.
- [8] K. Fukuda, S. Fujino, N. Mori, and K. Matsumoto, "Semi-automatic picture book generation based on story model and agent-based simulation," in *Intelligent and Evolutionary Systems: The 20th Asia Pacific Symposium, IES 2016, Canberra, Australia, November 2016, Proceedings*. Springer, 2017, pp. 117–132.
- [9] Y. Zhu, R. Kiros, R. S. Zemel, R. Salakhutdinov, R. Urtasun, A. Torralba, and S. Fidler, "Aligning books and movies: Towards story-like visual explanations by watching movies and reading books," in *2015 IEEE International Conference on Computer Vision, ICCV 2015, Santiago, Chile, December 7-13, 2015*. IEEE Computer Society, 2015, pp. 19–27. [Online]. Available: <https://doi.org/10.1109/ICCV.2015.11>
- [10] E. Nichols, L. Gao, Y. Vasyukiv, and R. Gomez, "Design and analysis of a collaborative story generation game for social robots," *Frontiers in Computer Science*, vol. 3, 2021. [Online]. Available: <https://www.frontiersin.org/articles/10.3389/fcomp.2021.674333>
- [11] D. Wilmot and F. Keller, "A temporal variational model for story generation," *ArXiv*, vol. abs/2109.06807, 2021. [Online]. Available: <https://api.semanticscholar.org/CorpusID:237503438>
- [12] K. Gregor, G. Papamakarios, F. Besse, L. Buesing, and T. Weber, "Temporal difference variational auto-encoder," *arXiv preprint arXiv:1806.03107*, 2018.
- [13] J. Guan, Z. Zhang, Z. Feng, Z. Liu, W. Ding, X. Mao, C. Fan, and M. Huang, "OpenMEVA: A benchmark for evaluating open-ended story generation metrics," in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, C. Zong, F. Xia, W. Li, and R. Navigli, Eds. Online: Association for Computational Linguistics, Aug. 2021, pp. 6394–6407. [Online]. Available: <https://aclanthology.org/2021.acl-long.500>
- [14] Z. Xie, T. Cohn, and J. H. Lau, "The next chapter: A study of large language models in storytelling," in *Proceedings of the 16th International Natural Language Generation Conference*, C. M. Keet, H.-Y. Lee, and S. Zarri , Eds. Prague, Czechia: Association for Computational Linguistics, Sep. 2023, pp. 323–351. [Online]. Available: <https://aclanthology.org/2023.inlg-main.23>
- [15] J. Li, M. Galley, C. Brockett, J. Gao, and B. Dolan, "A diversity-promoting objective function for neural conversation models," in *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, K. Knight, A. Nenkova, and O. Rambow, Eds. San Diego, California: Association for Computational Linguistics, Jun. 2016, pp. 110–119. [Online]. Available: <https://aclanthology.org/N16-1014>
- [16] Y. Zhu, S. Lu, L. Zheng, J. Guo, W. Zhang, J. Wang, and Y. Yu, "Texygen: A benchmarking platform for text generation models," *SIGIR*, 2018.
- [17] W. Yuan, G. Neubig, and P. Liu, "Bartscore: Evaluating generated text as text generation," in *Advances in Neural Information Processing Systems*, M. Ranzato, A. Beygelzimer, Y. Dauphin, P. Liang, and J. W. Vaughan, Eds., vol. 34. Curran Associates, Inc., 2021, pp. 27 263–27 277. [Online]. Available: <https://proceedings.neurips.cc/paper/2021/file/e4d2b6e6fdeca3e60e0f1a62fee3d9dd-Paper.pdf>
- [18] Z. Shao, M. Huang, J. Wen, W. Xu, and X. Zhu, "Long and diverse text generation with planning-based hierarchical variational model," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, K. Inui, J. Jiang, V. Ng, and X. Wan, Eds. Hong Kong, China: Association for Computational Linguistics, Nov. 2019, pp. 3257–3268. [Online]. Available: <https://aclanthology.org/D19-1321>
- [19] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozi re, N. Goyal, E. Hambro, F. Azhar *et al.*, "Llama: Open and efficient foundation language models," *arXiv preprint arXiv:2302.13971*, 2023.
- [20] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. Henighan, R. Child, A. Ramesh, D. M. Ziegler, J. Wu, C. Winter, C. Hesse, M. Chen, E. Sigler, M. Litwin, S. Gray, B. Chess, J. Clark, C. Berner, S. McCandlish, A. Radford, I. Sutskever, and D. Amodei, "Language models are few-shot learners," 2020.
- [21] H. Touvron, L. Martin, K. Stone, P. Albert, A. Almahairi, Y. Babaei, N. Bashlykov, S. Batra, P. Bhargava, S. Bhosale *et al.*, "Llama 2: Open foundation and fine-tuned chat models," *arXiv preprint arXiv:2307.09288*, 2023.
- [22] X.-L. Team, "Xwin-lm," 9 2023. [Online]. Available: <https://github.com/Xwin-LM/Xwin-LM>
- [23] X. Li, T. Zhang, Y. Dubois, R. Taori, I. Gulrajani, C. Guestrin, P. Liang, and T. B. Hashimoto, "AlpacaEval: An automatic evaluator of instruction-following models," https://github.com/tatsu-lab/alpaca_eval, 2023.
- [24] Y. Dubois, X. Li, R. Taori, T. Zhang, I. Gulrajani, J. Ba, C. Guestrin, P. Liang, and T. B. Hashimoto, "AlpacaFarm: A simulation framework for methods that learn from human feedback," 2023.
- [25] OpenAI, "Gpt-4 technical report," 2023.
- [26] Z. M. Wang, Z. Peng, H. Que, J. Liu, W. Zhou, Y. Wu, H. Guo, R. Gan, Z. Ni, M. Zhang, Z. Zhang, W. Ouyang, K. Xu, W. Chen, J. Fu, and J. Peng, "Rolellm: Benchmarking, eliciting, and enhancing role-playing abilities of large language models," *arXiv preprint arXiv: 2310.00746*, 2023.
- [27] Y. Shao, L. Li, J. Dai, and X. Qiu, "Character-llm: A trainable agent for role-playing," in *EMNLP*, 2023.
- [28] J. E. Hu, R. Rudinger, M. Post, and B. Van Durme, "Parabank: Monolingual bitext generation and sentential paraphrasing via lexically-constrained neural machine translation," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, no. 01, 2019, pp. 6521–6528.
- [29] N. Mostafazadeh, N. Chambers, X. He, D. Parikh, D. Batra, L. Vanderwende, P. Kohli, and J. Allen, "A corpus and cloze evaluation for deeper understanding of commonsense stories," in *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, K. Knight, A. Nenkova, and O. Rambow, Eds. San Diego, California: Association for Computational Linguistics, Jun. 2016, pp. 839–849. [Online]. Available: <https://aclanthology.org/N16-1098>
- [30] A. Fan, M. Lewis, and Y. Dauphin, "Hierarchical neural story generation," in *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, I. Gurevych and Y. Miyao, Eds. Melbourne, Australia: Association for Computational Linguistics, Jul. 2018, pp. 889–898. [Online]. Available: <https://aclanthology.org/P18-1082>
- [31] K. M. Hermann, T. Kocisky, E. Grefenstette, L. Espeholt, W. Kay, M. Suleyman, and P. Blunsom, "Teaching machines to read and comprehend," *Advances in neural information processing systems*, vol. 28, 2015.